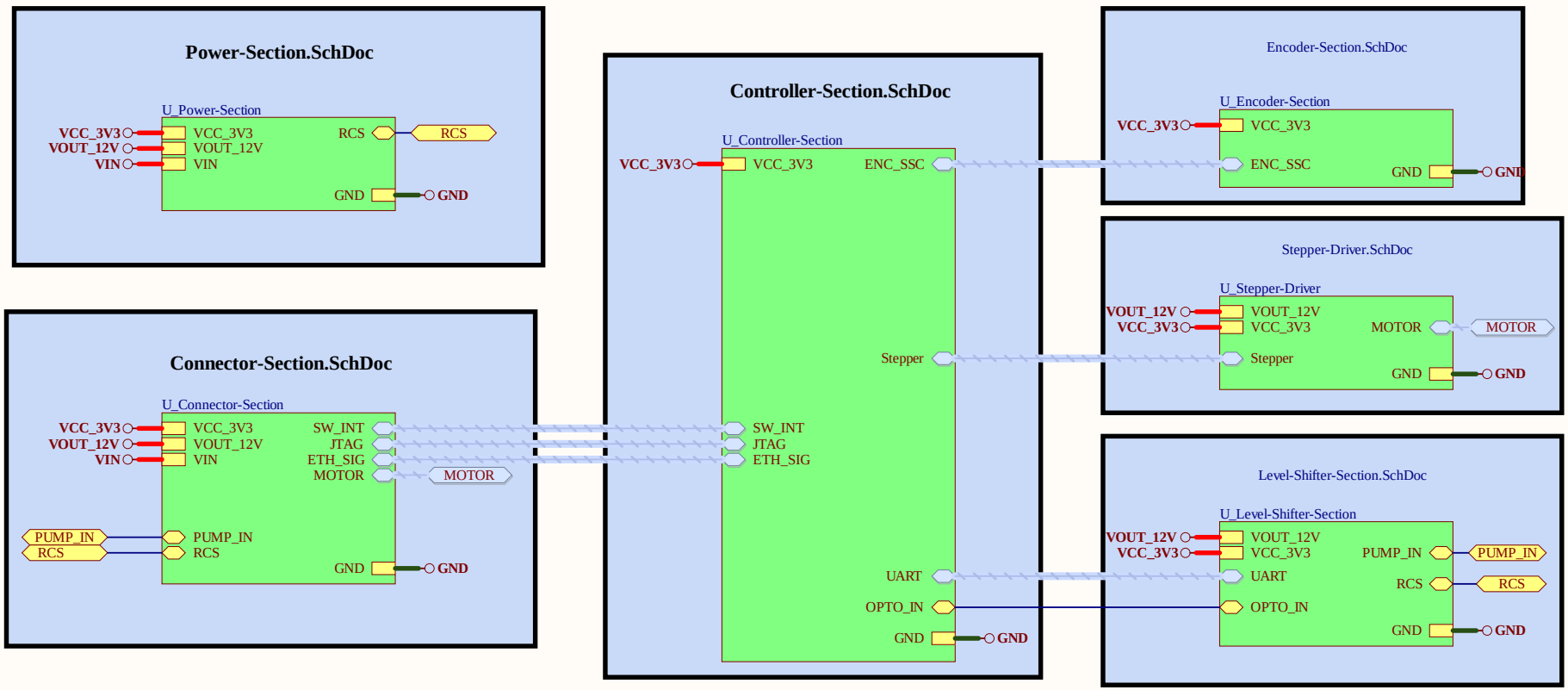
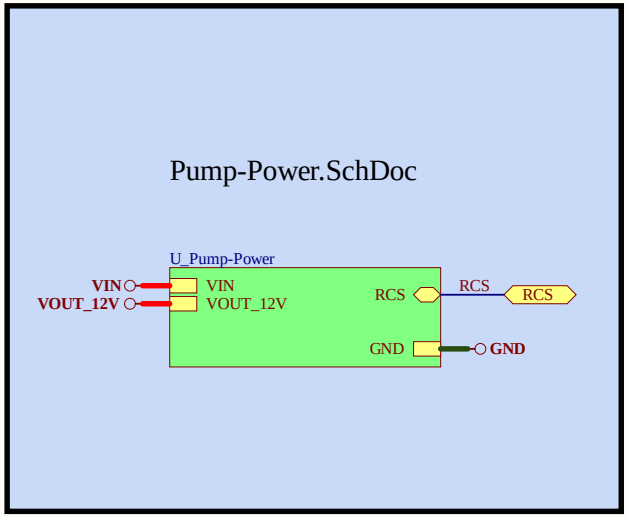
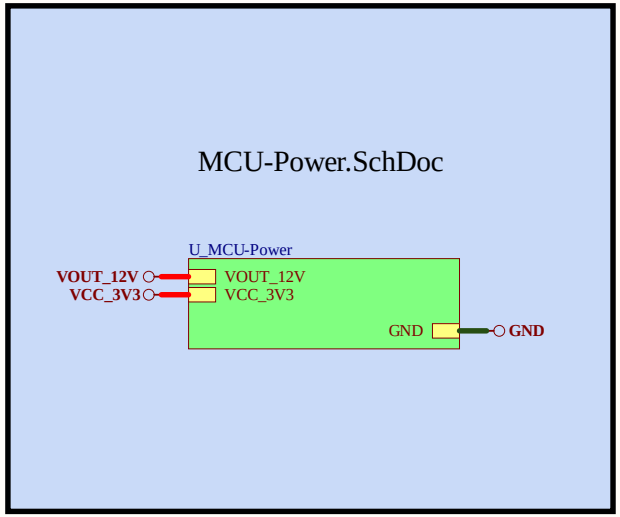
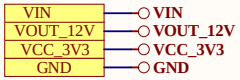
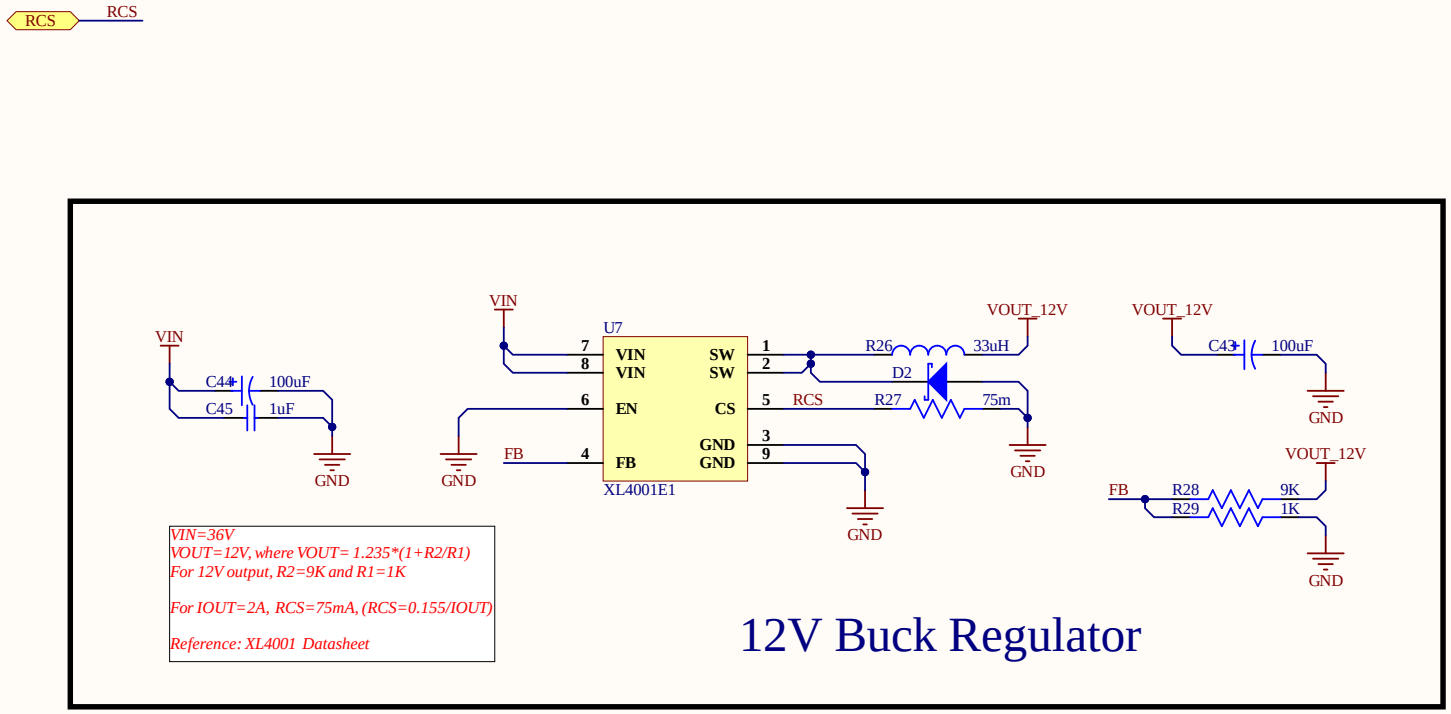
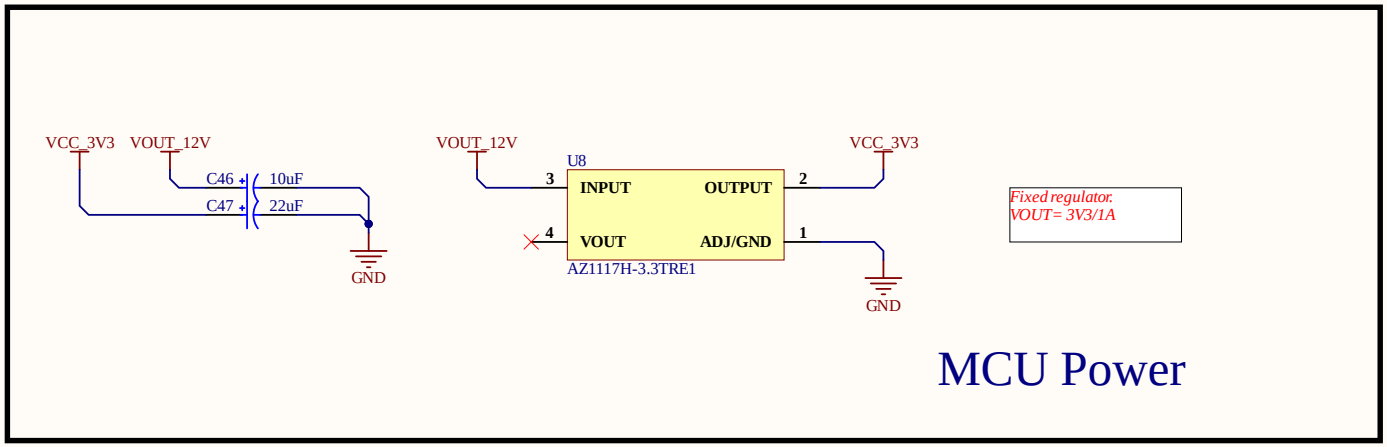

Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
VCC_3V3
VOUT_12V
VIN
RCS
GND
RCS
GND
Connector-Section.SchDoc
U_Connector-Section
VCC_3V3
VOUT_12V
VIN
PUMP_IN
RCS
VCC_3V3
VOUT_12V
VIN
SW_INT
JTAG
ETH_SIG
MOTOR
PUMP_IN
RCS
GND
MOTOR
GND
Controller-Section.SchDoc
U_Controller-Section
VCC_3V3
VCC_3V3
ENC_SSC
Stepper
SW_INT
JTAG
ETH_SIG
UART
OPTO_IN
GND
GND
Encoder-Section.SchDoc
U_Encoder-Section
VCC_3V3
VCC_3V3
ENC_SSC
GND
GND
Stepper-Driver.SchDoc
U_Stepper-Driver
VOUT_12V
VCC_3V3
VOUT_12V
VCC_3V3
MOTOR
Stepper
GND
MOTOR
GND
Level-Shifter-Section.SchDoc
U_Level-Shifter-Section
VOUT_12V
VCC_3V3
VOUT_12V
VCC_3V3
UART
OPTO_IN
PUMP_IN
RCS
GND
PUMP_IN
RCS
GND


Pump-Power.SchDoc
U_Pump-Power
VIN
VOUT_12V
VIN
VOUT_12V
RCS
RCS
RCS
GND
GND

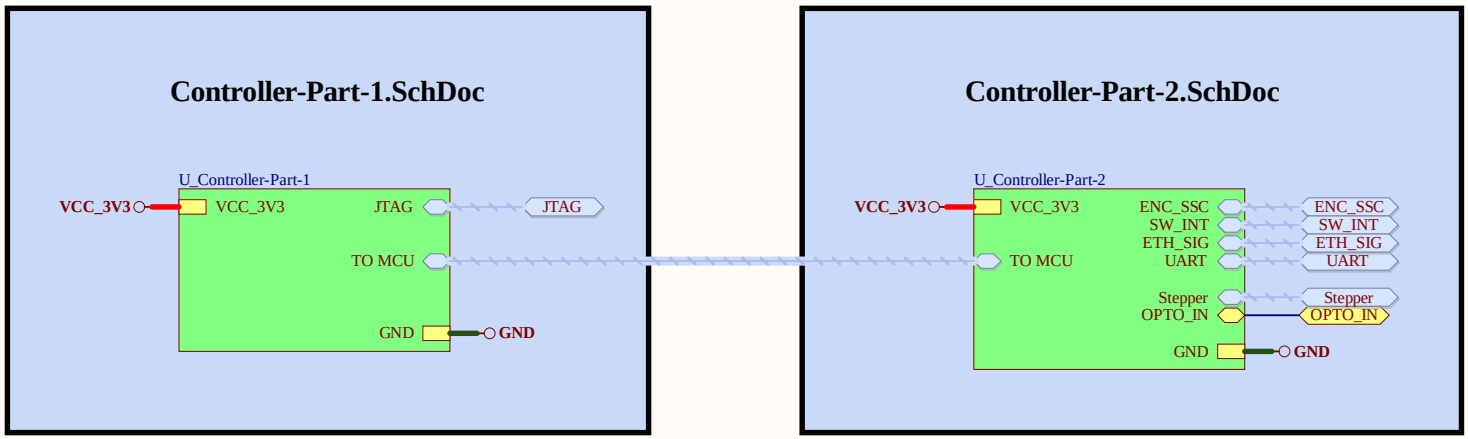

MCU-Power.SchDoc
U_MCU-Power
VOUT_12V
VCC_3V3
VOUT_12V
VCC_3V3
GND
GND

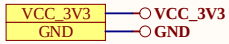

VIN
VOUT_12V
VCC_3V3
GND
VIN
VOUT_12V
VCC_3V3
GND

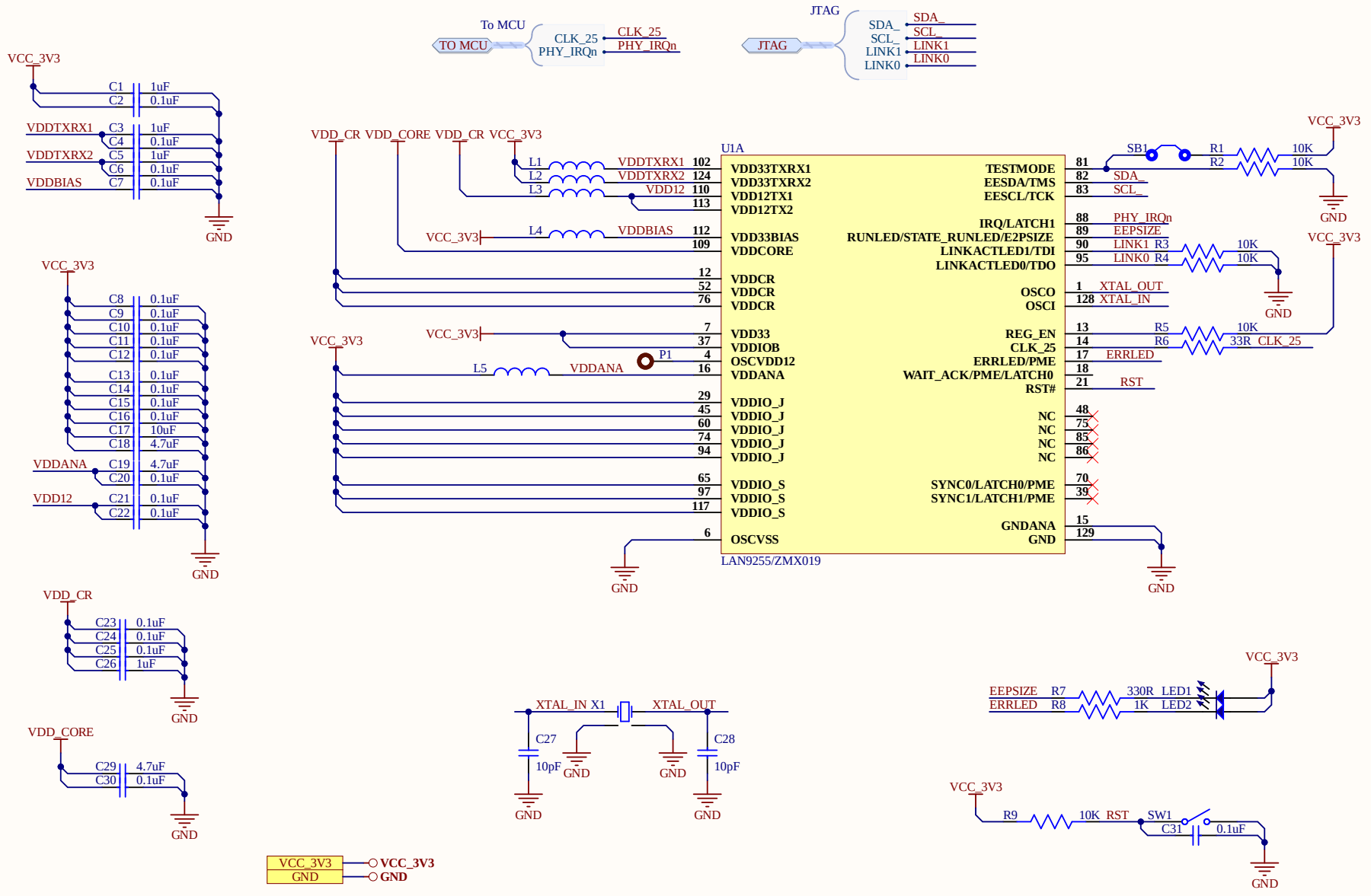

RCS
RCS
VIN
C44
100uF
C45
1uF
GND
VIN
U7
7
8
6
FB
4
VIN
VIN
EN
FB
SW
SW
CS
GND
GND
XL4001E1
1
2
5
3
9
RCS
R26
33uH
D2
R27
75m
VOUT_12V
GND
GND
GND
VOUT_12V
C43
100uF
GND
VOUT_12V
FB
R28
9K
R29
1K
GND
VIN=36V
VOUT=12V, where VOUT= 1.235*(1+R2/R1)
For 12V output, R2=9K and R1=1K
For IOUT=2A, RCS=75mA, (RCS=0.155/IOUT)
Reference: XL4001 Datasheet
12V Buck Regulator
VIN
VOUT_12V
GND
VIN
VOUT_12V
GND

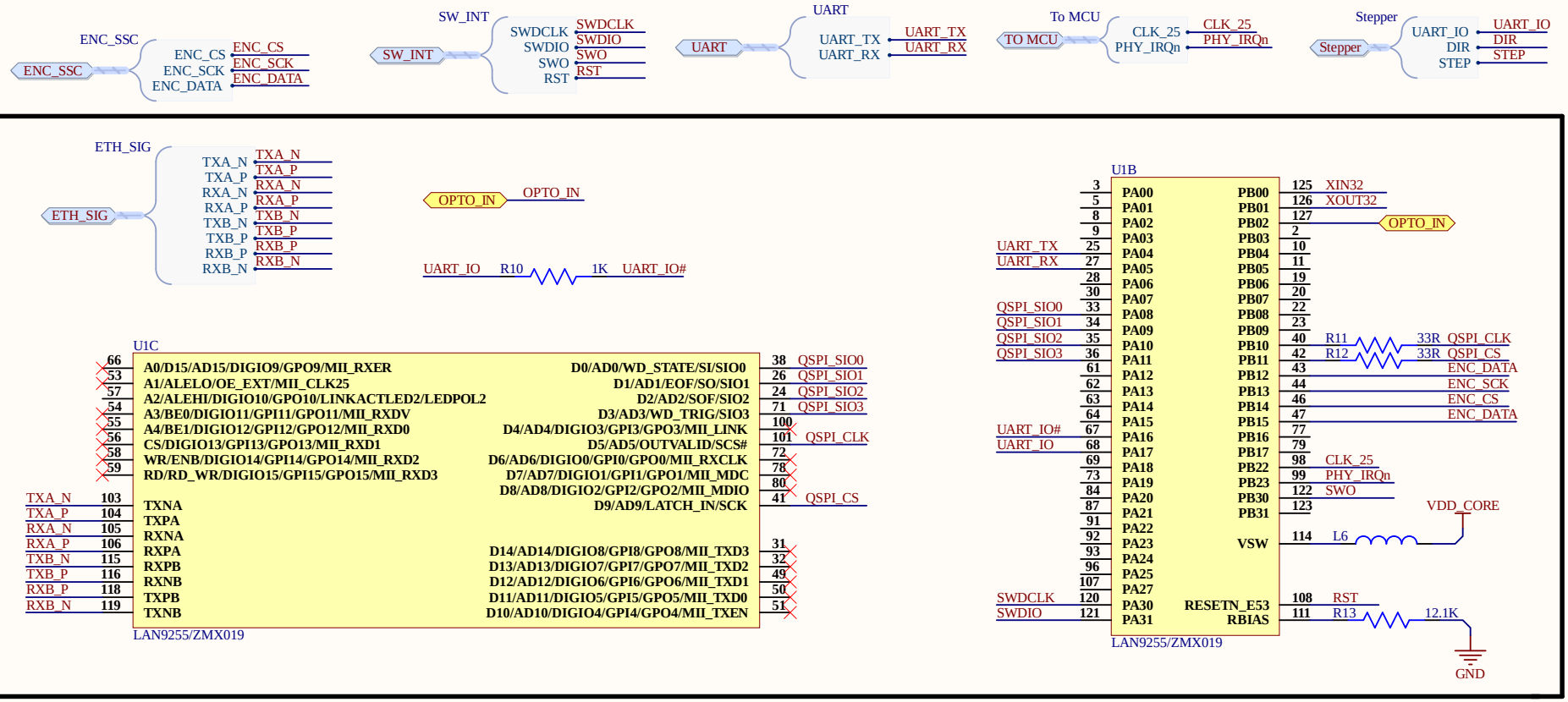


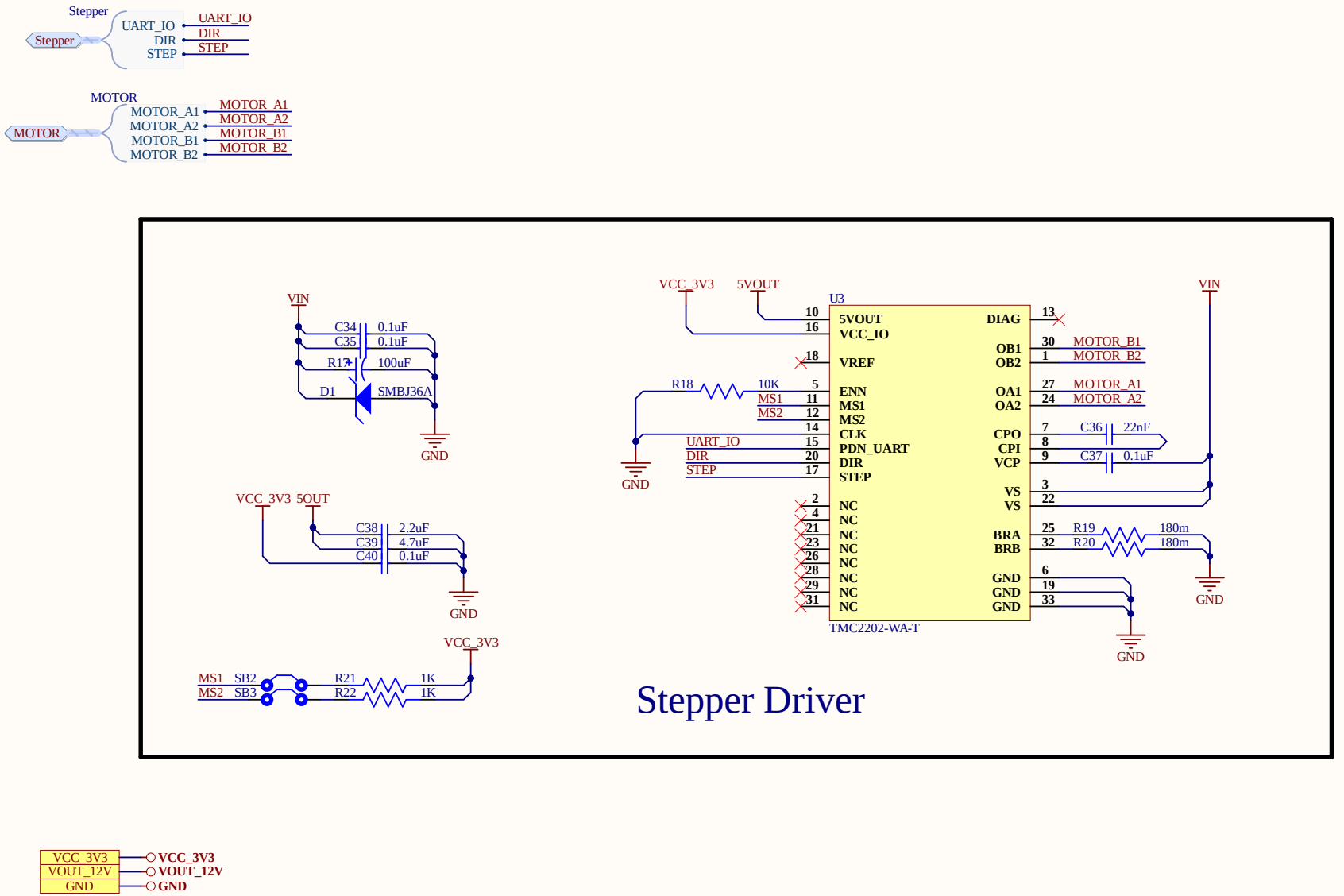
VOUT_12V — VOUT_12V
VCC_3V3 — VCC_3V3
GND — GND

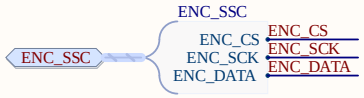

Controller-Part-1.SchDoc
U_Controller-Part-1
VCC_3V3
VCC_3V3
JTAG
JTAG
TO MCU
GND
GND
Controller-Part-2.SchDoc
U_Controller-Part-2
VCC_3V3
VCC_3V3
TO MCU
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
GND
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
GND

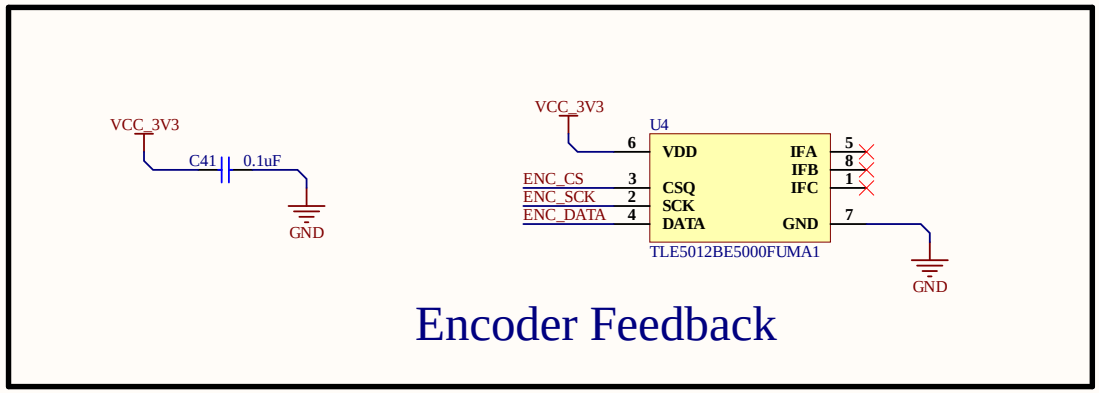

VCC_3V3
GND
VCC_3V3
GND

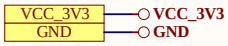

To MCU
TO MCU
CLK_25
PHY_IRQn
CLK_25
PHY_IRQn
JTAG
JTAG
SDA_
SCL_
LINK1
LINK0
SDA_
SCL_
LINK1
LINK0
U1A
LAN9255/ZMX019
VDD33TXRX1
VDD33TXRX2
VDD12TX1
VDD12TX2
VDD33BIAS
VDDCORE
VDDCR
VDDCR
VDDCR
VDD33
VDDIOB
OSCVDD12
VDDANA
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_S
VDDIO_S
VDDIO_S
OSCVSS
TESTMODE
EESDA/TMS
EESCL/TCK
IRQ/LATCH1
RUNLED/STATE_RUNLED/E2PSIZE
LINKACTLED1/TDI
LINKACTLED0/TDO
OSCO
OSCI
REG_EN
CLK_25
ERRLED/PME
WAIT_ACK/PME/LATCH0
RST#
NC
NC
NC
NC
SYNC0/LATCH0/PME
SYNC1/LATCH1/PME
GNDANA
GND
SB1
R1
R2
10K
SDA_
SCL_
PHY_IRQn
EEPSIZE
LINK1 R3
LINK0 R4
XTAL_OUT
XTAL_IN
R5
R6
33R CLK_25
ERRLED
RST
VCC_3V3
GND
VDD_CR VDD_CORE VDD_CR VCC_3V3
L1 VDDTXRX1
L2 VDDTXRX2
L3 VDD12
L4 VDDBIAS
L5 VDDANA
P1
C1 1uF
C2 0.1uF
C3 1uF
C4 0.1uF
C5 1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
C12 0.1uF
C13 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 10uF
C18 4.7uF
C19 4.7uF
C20 0.1uF
C21 0.1uF
C22 0.1uF
C23 0.1uF
C24 0.1uF
C25 0.1uF
C26 1uF
C27 10pF
C28 10pF
C29 4.7uF
C30 0.1uF
C31 0.1uF
XTAL_IN X1
XTAL_OUT
EEPSIZE R7 330R LED1
ERRLED R8 1K LED2
R9 10K RST SW1
VCC_3V3
GND
VCC_3V3
GND


ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
SW_INT
SWDCLK
SWDIO
SWO
RST
UART
UART_TX
UART_RX
To MCU
CLK_25
PHY_IRQn
Stepper
UART_IO
DIR
STEP
ETH_SIG
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_P
RXB_N
OPTO_IN
UART_IO
R10
1K
UART_IO#
U1C
A0/D15/AD15/DIGIO9/GPO9/MII_RXER
A1/ALELO/OE_EXT/MII_CLK25
A2/ALEHI/DIGIO10/GPO10/LINKACTLED2/LEDPOL2
A3/BE0/DIGIO11/GPI11/GPO11/MII_RXDV
A4/BE1/DIGIO12/GPI12/GPO12/MII_RXD0
CS/DIGIO13/GPI13/GPO13/MII_RXD1
WR/ENB/DIGIO14/GPI14/GPO14/MII_RXD2
RD/RD_WR/DIGIO15/GPI15/GPO15/MII_RXD3
TXNA
TXPA
RXNA
RXPA
RXPB
RXNB
TXPB
TXNB
D0/AD0/WD_STATE/SI/SIO0
D1/AD1/EOF/SO/SIO1
D2/AD2/SOF/SIO2
D3/AD3/WD_TRIG/SIO3
D4/AD4/DIGIO3/GPI3/GPO3/MII_LINK
D5/AD5/OUTVALID/SCS#
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D9/AD9/LATCH_IN/SCK
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3
D13/AD13/DIGIO7/GPI7/GPO7/MII_TXD2
D12/AD12/DIGIO6/GPI6/GPO6/MII_TXD1
D11/AD11/DIGIO5/GPI5/GPO5/MII_TXD0
D10/AD10/DIGIO4/GPI4/GPO4/MII_TXEN
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_CLK
QSPI_CS
LAN9255/ZMX019
U1B
PA00
PA01
PA02
PA03
PA04
PA05
PA06
PA07
PA08
PA09
PA10
PA11
PA12
PA13
PA14
PA15
PA16
PA17
PA18
PA19
PA20
PA21
PA22
PA23
PA24
PA25
PA27
PA30
PA31
PB00
PB01
PB02
PB03
PB04
PB05
PB06
PB07
PB08
PB09
PB10
PB11
PB12
PB13
PB14
PB15
PB16
PB17
PB22
PB23
PB30
PB31
VSW
RESETN_E53
RBIAS
XIN32
XOUT32
R11
R12
33R
ENC_DATA
ENC_SCK
ENC_CS
CLK_25
PHY_IRQn
SWO
VDD_CORE
L6
R13
12.1K
GND
VCC_3V3
R15
R16
10K
SDA_
SCL_
U2
VCC
WP
SDA
SCL
A0
A1
A2
VSS
R14
CAT24C256WI-GT3
EEPROM
X2
C32
C33
10pF
32.768KHz Crystal

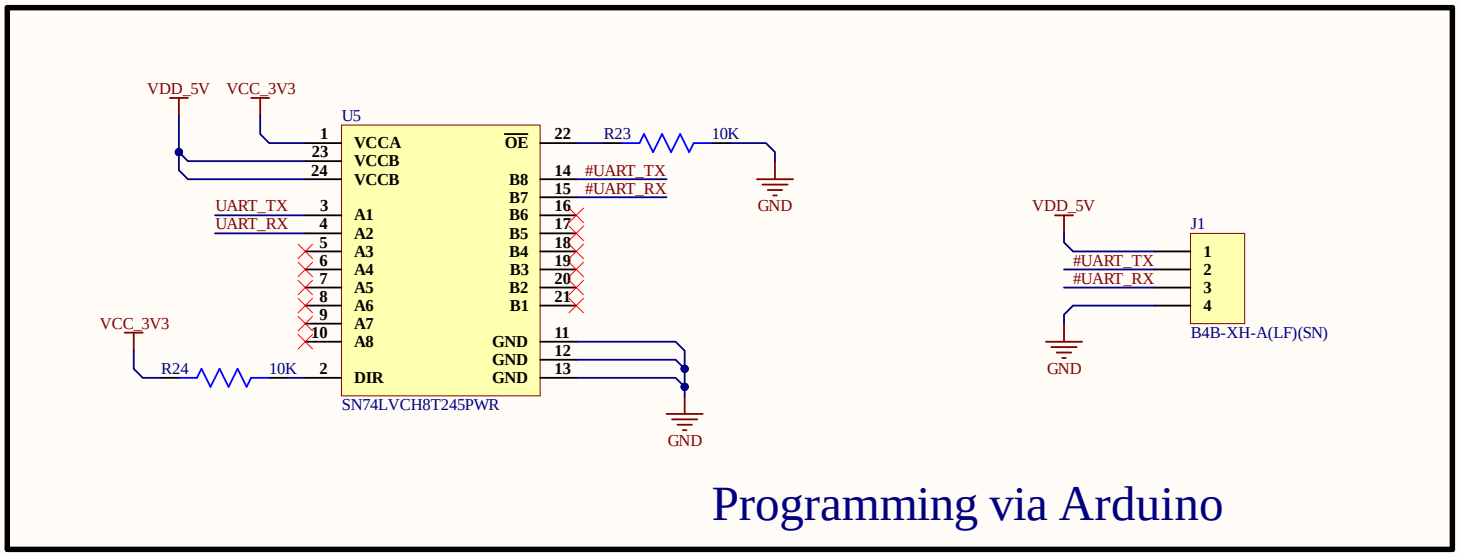

Stepper
Stepper
UART_IO
DIR
STEP
UART_IO
DIR
STEP
MOTOR
MOTOR
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
VIN
C34 0.1uF
C35 0.1uF
R17 100uF
D1 SMBJ36A
GND
VCC_3V3 5OUT
C38 2.2uF
C39 4.7uF
C40 0.1uF
GND
VCC_3V3
MS1 SB2
MS2 SB3
R21 1K
R22 1K
VCC_3V3 5VOUT
U3
10 5VOUT
16 VCC_IO
18 VREF
5 ENN
11 MS1
12 MS2
14 CLK
15 PDN_UART
20 DIR
17 STEP
2 NC
4 NC
21 NC
23 NC
26 NC
28 NC
29 NC
31 NC
TMC2202-WA-T
13 DIAG
30 OB1
1 OB2
27 OA1
24 OA2
7 CPO
8 CPI
9 VCP
3 VS
22 VS
25 BRA
32 BRB
6 GND
19 GND
33 GND
R18 10K
MS1
MS2
UART_IO
DIR
STEP
GND
C36 22nF
C37 0.1uF
VIN
R19 180m
R20 180m
GND
GND
Stepper Driver
VCC_3V3
VOUT_12V
GND
VCC_3V3
VOUT_12V
GND


ENC_SSC
ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
ENC_CS
ENC_SCK
ENC_DATA

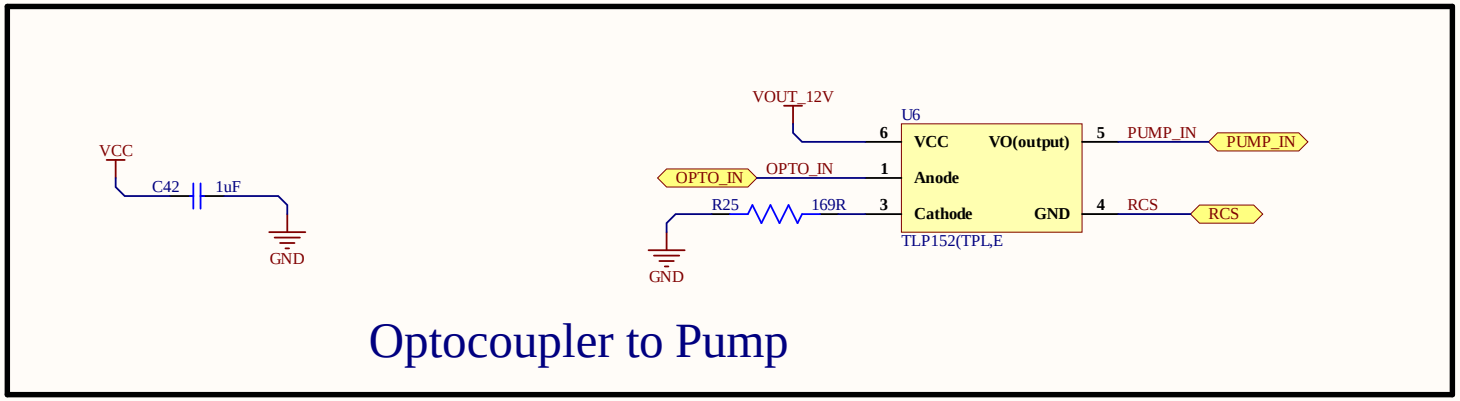

VCC_3V3
C41
0.1uF
GND
VCC_3V3
U4
6
VDD
IFA
5
8
IFB
1
IFC
ENC_CS
3
CSQ
ENC_SCK
2
SCK
ENC_DATA
4
DATA
GND
7
TLE5012BE5000FUMA1
GND
Encoder Feedback

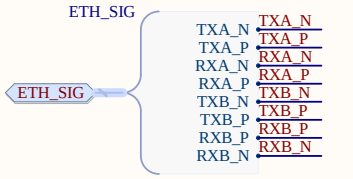

VCC_3V3
GND
VCC_3V3
GND

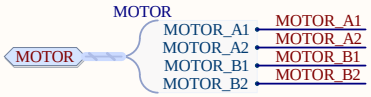

UART
UART
UART_TX
UART_RX
UART_TX
UART_RX

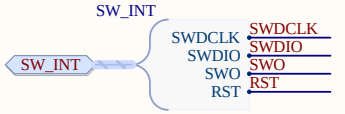

VDD_5V
VCC_3V3
U5
1
23
24
VCCA
VCCB
VCCB
OE
22
R23
10K
GND
B8
B7
B6
B5
B4
B3
B2
B1
14
15
16
17
18
19
20
21
#UART_TX
#UART_RX
UART_TX
UART_RX
3
4
A1
A2
A3
A4
A5
A6
A7
A8
5
6
7
8
9
10
VCC_3V3
R24
10K
2
DIR
GND
GND
GND
SN74LVCH8T245PWR
VDD_5V
J1
#UART_TX
#UART_RX
1
2
3
4
B4B-XH-A(LF)(SN)
GND
Programming via Arduino


VOUT_12V
VCC_3V3
GND
VOUT_12V
VCC_3V3
GND

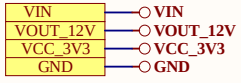

VCC
C42
1uF
GND
VOUT_12V
U6
6
VCC
VO(output)
5
PUMP_IN
PUMP_IN
OPTO_IN
OPTO_IN
1
Anode
R25
169R
3
Cathode
GND
4
RCS
RCS
GND
TLP152(TPL,E
Optocoupler to Pump

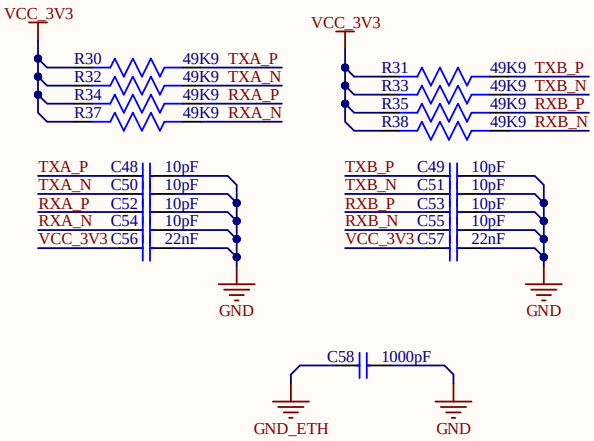


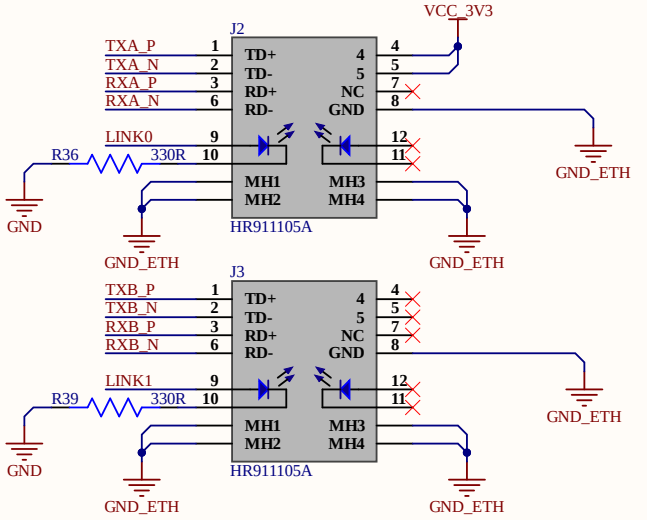




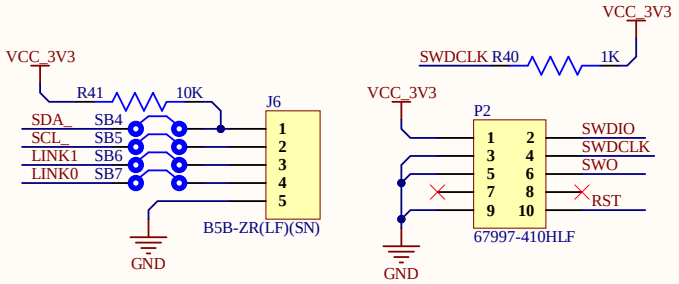




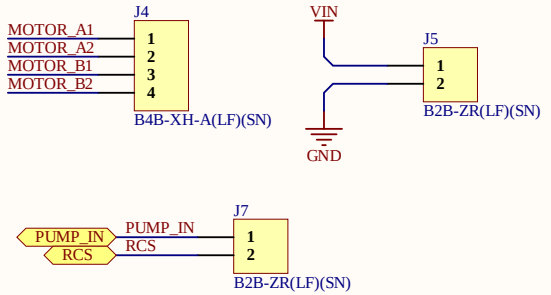




EtherCAT Input



Programming Headers



Power In and Out